Blackmon Lab

Evolutionary Biology

Desert Specialization Does Not Drive North American *Astragalus* Radiation

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Background

The hyper-diversification of *Astragalus* in North American deserts presents a fundamental issue. Foundational theory (Stebbins 1950, 1971; Cohen 1966) suggests a causal chain for desert colonization:

- Descending Dysploidy: Physically restricts recombination to "lock in" complex drought-survival traits.
- Life-History: This genomic stabilization forces a shift to annuality as a "drought-escape" strategy.

While ancestral Eurasian lineages remain anchored to a stable diploid baseline (n=8), the North American radiation defies this model:

- Chromosome Explosion: North American clades exhibit a broad range of chromosome counts (n=11–45) in hyper-arid niches.
- The Gap: Early tests (Sanderson 1991) questioned this link but lacked deep-time phylogenomic resolution and high-resolution spatial modeling required for a definitive test.

To resolve this discrepancy, I built a high-resolution computational pipeline:

1. Phylogenomics: 318-species global phylogeny anchored by the Eurasian control.
2. Bayesian Modeling: ChromeSSE (MCMC) in Revbayes to track the rate and direction of chromosomal mutations
3. Spatial Ecology: Thousands of GBIF occurrences mapped via WorldClim 2.1 to define precipitation (bio_12) and seasonality (bio_15) niches.

Therefore, this study seeks to answer: **does chromosomal instability facilitate colonization of novel environmental niches, or does it function as a mechanism for high-volume diversification within a converged ancestral climate?**

Study Design & AI

I used AI to help me discover novel research topics involving the clade *Fabaceae*, which I then used to narrow down a specific question I wanted to research and answer. The AI then helped me write code in order to run various tests in order to answer my question; requiring that I checked between each step to ensure quality and accuracy with what I was trying to achieve. Ultimately, I was able to come to a conclusion through analyzing my findings.

Acknowledgements

Thank you to Dr. Blackmon and the Blackmon Lab for his mentorship throughout the entire research process with the CUREs.

Reversing the Evolutionary Engine

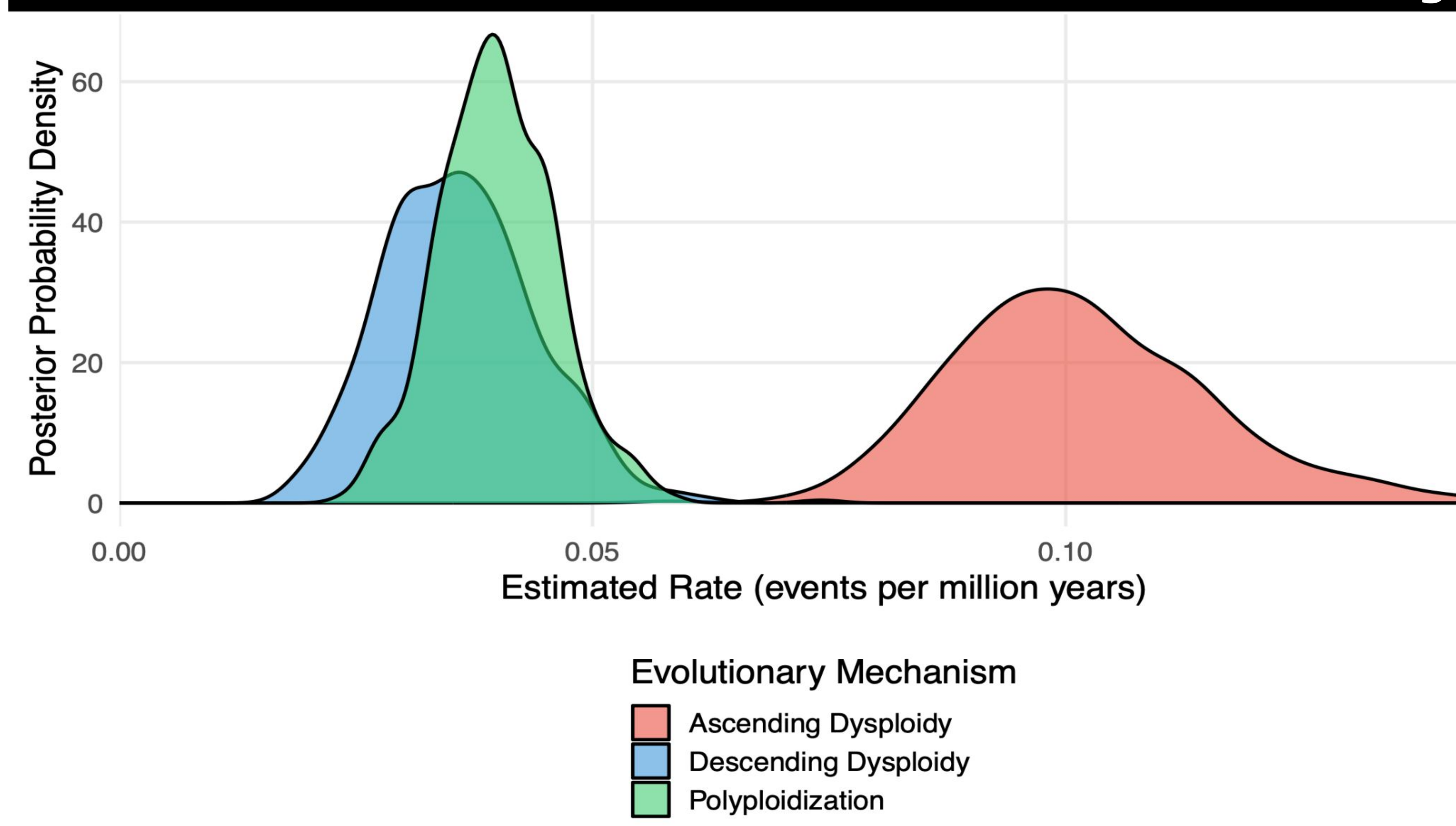

Figure 1. Bayesian posterior probability densities of chromosome evolution.

Figure 1:

- Ascending Dysploidy (chromosome gain) is the primary driver of *Astragalus* evolution, occurring at a significantly higher mean rate (0.1013 events/Myr) than Descending Dysploidy (0.0360 events/Myr) or polyploidization (0.0399 events/Myr). This fundamentally rejects classical assumptions of genome "shrinking" for survival.

Figure 2:

- Ancestral states reveal a stable diploid baseline (n ≤ 8) in Eurasia. Upon radiation into North America, the karyotype transitioned into extreme instability (dysploidy: n = 11-15; polyploidy: n ≥1 6)
- Despite this chromosomal "explosion," continuous climate mapping illustrates that North American lineages occupy environmental conditions nearly identical to their ancestors.

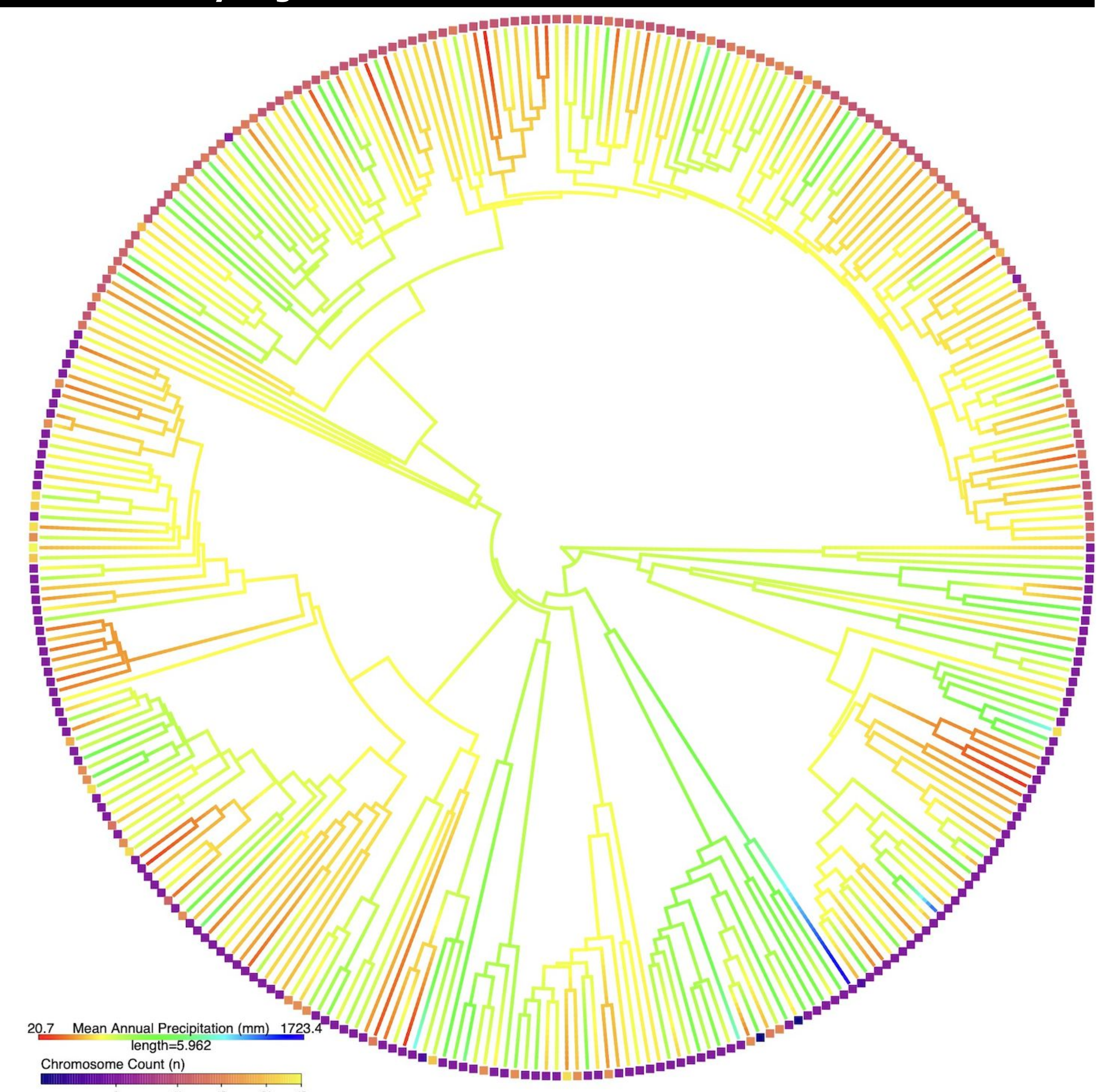

Figure 2. Ancestral state reconstruction of Mean Annual Precipitation across 318 species

The Ecological Correlates

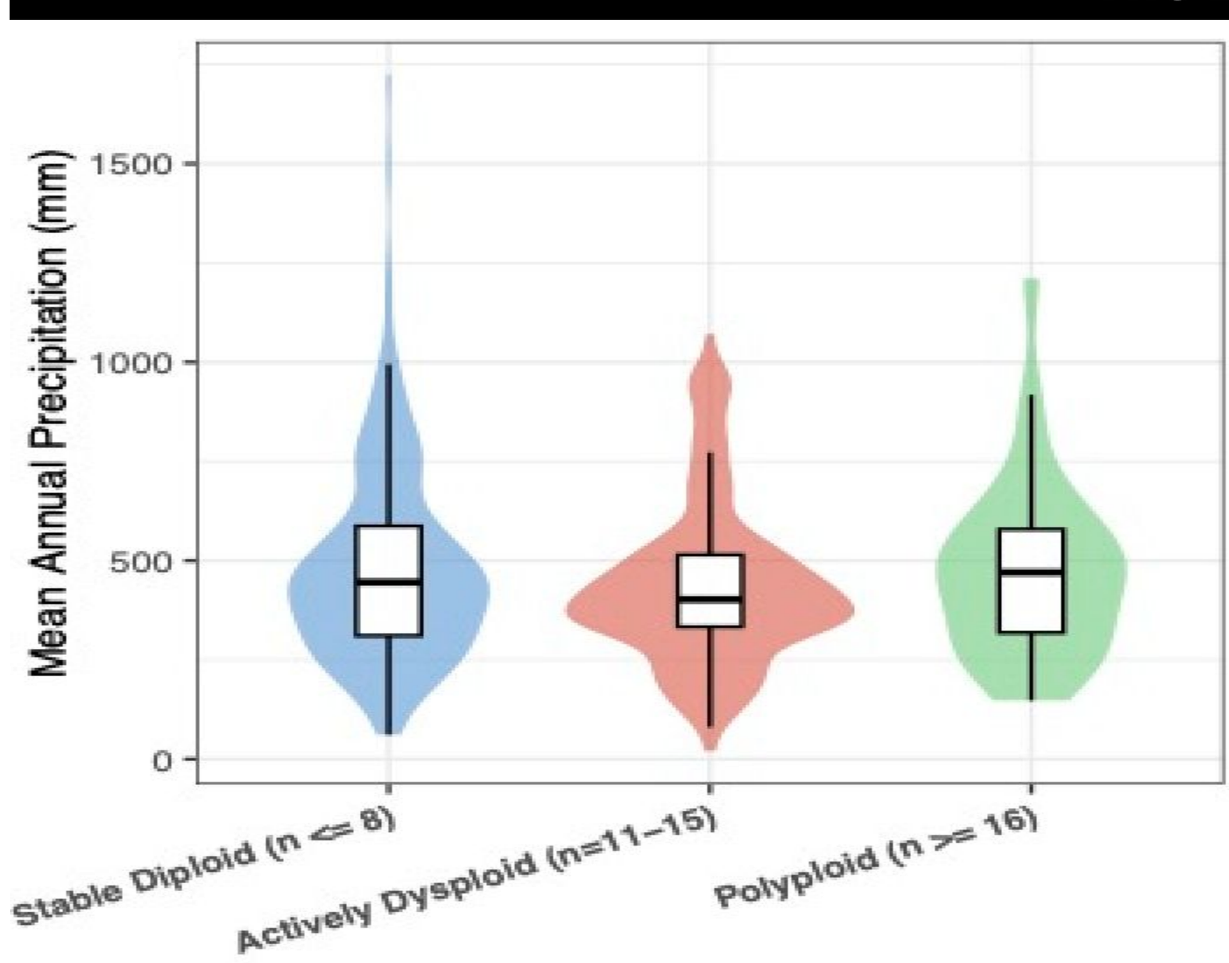

Figure 3. Niche breadth analysis across cytogenetic groups

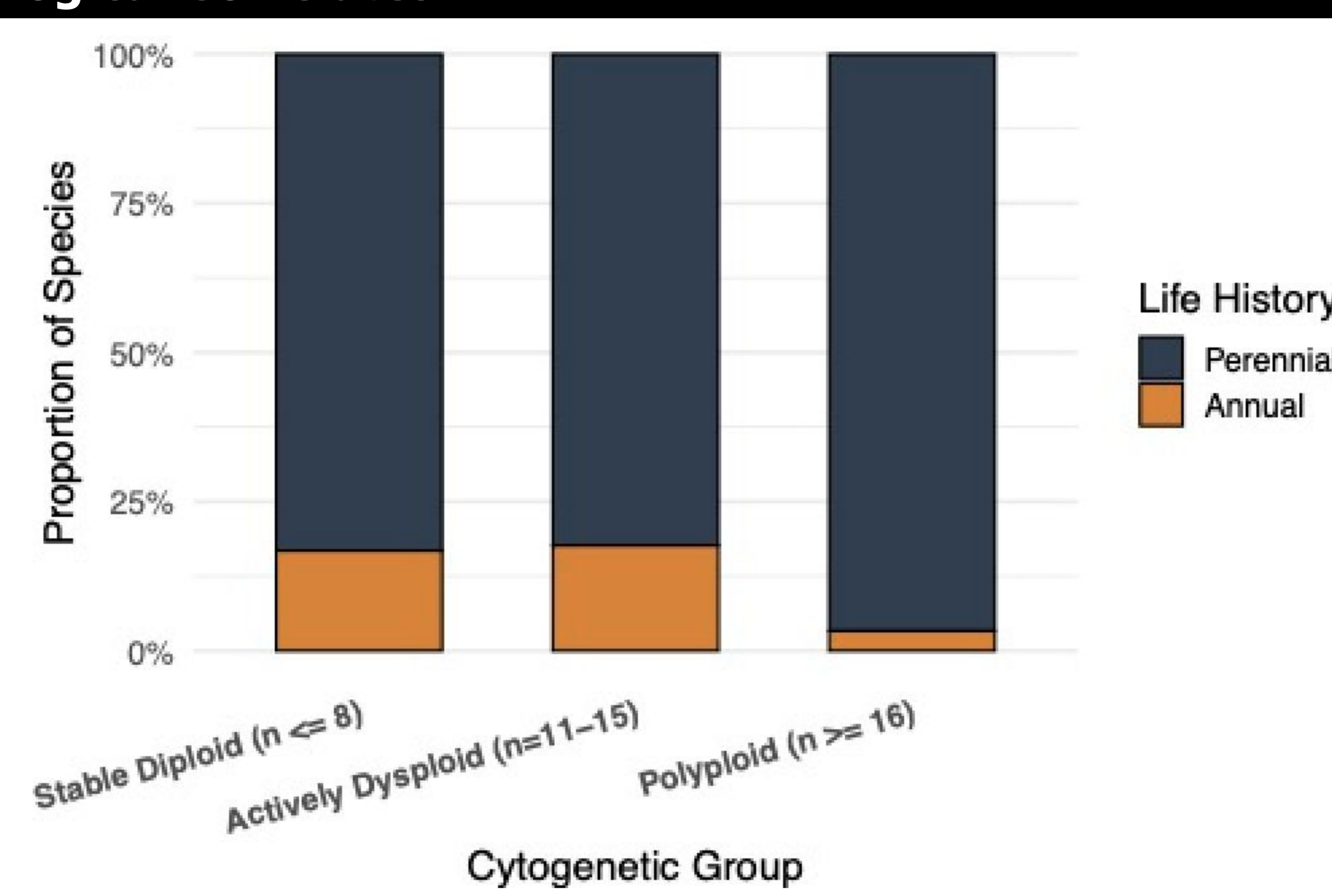

Figure 4. Life history strategy analysis across cytogenetic groups

Figure 3 & 4:

- The violin distributions in **Figure 3** illustrate that Stable Diploids, Actively Disploids, and Polyploid lineages occupy statistically identical precipitation niches (median ≈ 450mm). Phylogenetic ANOVAs ($p = 0.908$) confirm that chromosomal instability does not drive colonization of drier ecological niches, demonstrating robust niche conservatism across the radiation.
- The stacked proportions in **Figure 4** illustrates that chromosomal instability does not trigger the predicted transition to annual "desert escape" strategy. Phylogenetic logistic regression ($p = 0.455$) confirms that life history transitions are decoupled from karyotype evolution; the North American radiation remains overwhelmingly perennial

Conclusion

- We have completed phase 1 of this study which was designed to determine whether the data distribution is sufficient to use this as a model group for studying the impacts of niche evolution on genome evolution.
- Based on our findings we believe Astragalus is an ideal system for studying these questions:
  - informed estimates of rates
  - broad distribution of species across habitats
- We plan on doing a follow on study that will explicitly test associations between invasion of dessert habitats and rates of chromosome evolution.